



Computer Communication Laboratory Seminar

OIRS Element Allocation for Integrated FSO-based HAP and sub-THz Access Networks

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Aizuwakamatsu, May 20, 2026

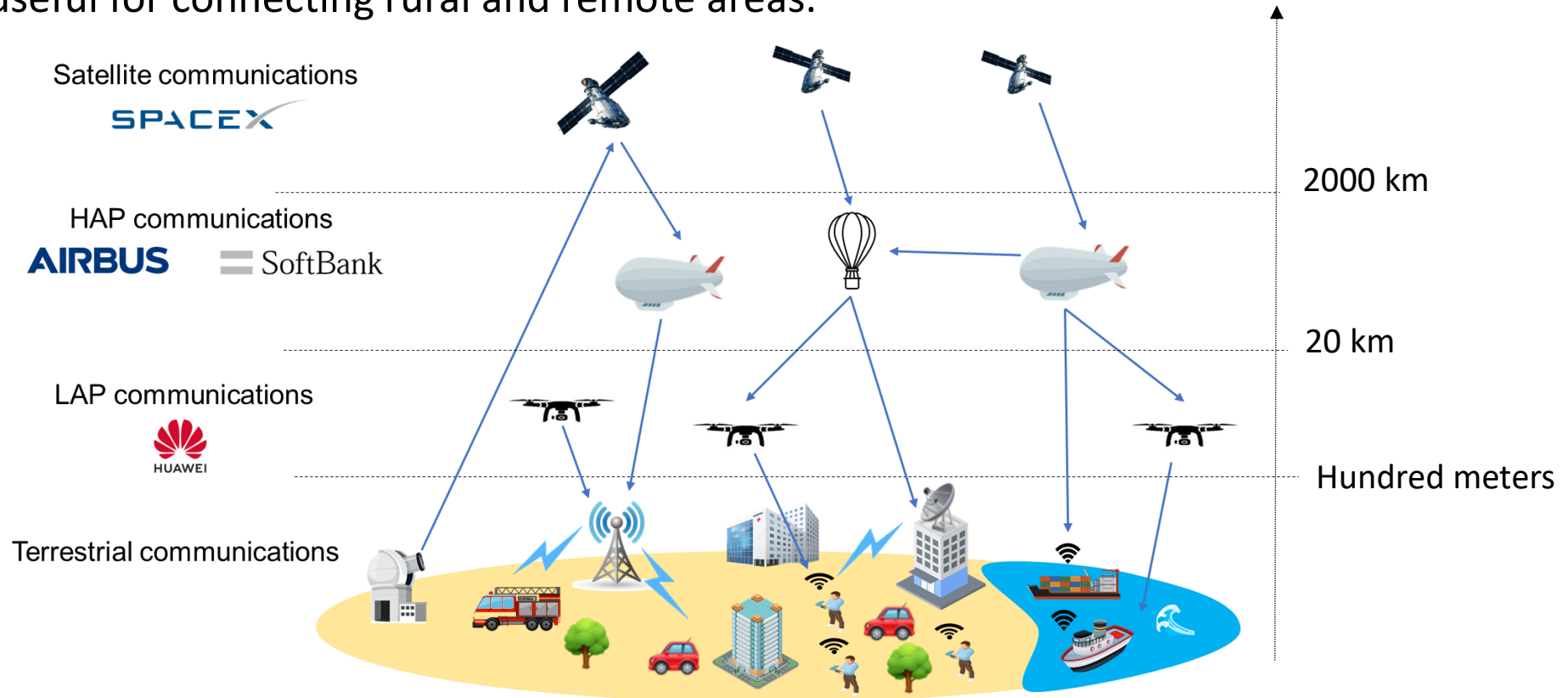
Outline of Presentation

I. Review of OIRS allocation in the Integrated Network

II. Research proposal

Non-Terrestrial Networks

- ❑ Future 6G networks are expected to provide **ubiquitous connectivity** and **ultra high-data rate**, even in remote/rural/hard-to-reach areas.
- ❑ Non-terrestrial networks (NTNs) can **provide internet from space** by using *satellites, HAPs and UAVs*, especially useful for connecting rural and remote areas.



- ❑ Several successful projects have demonstrated the **feasibility and potential** of NTNs.

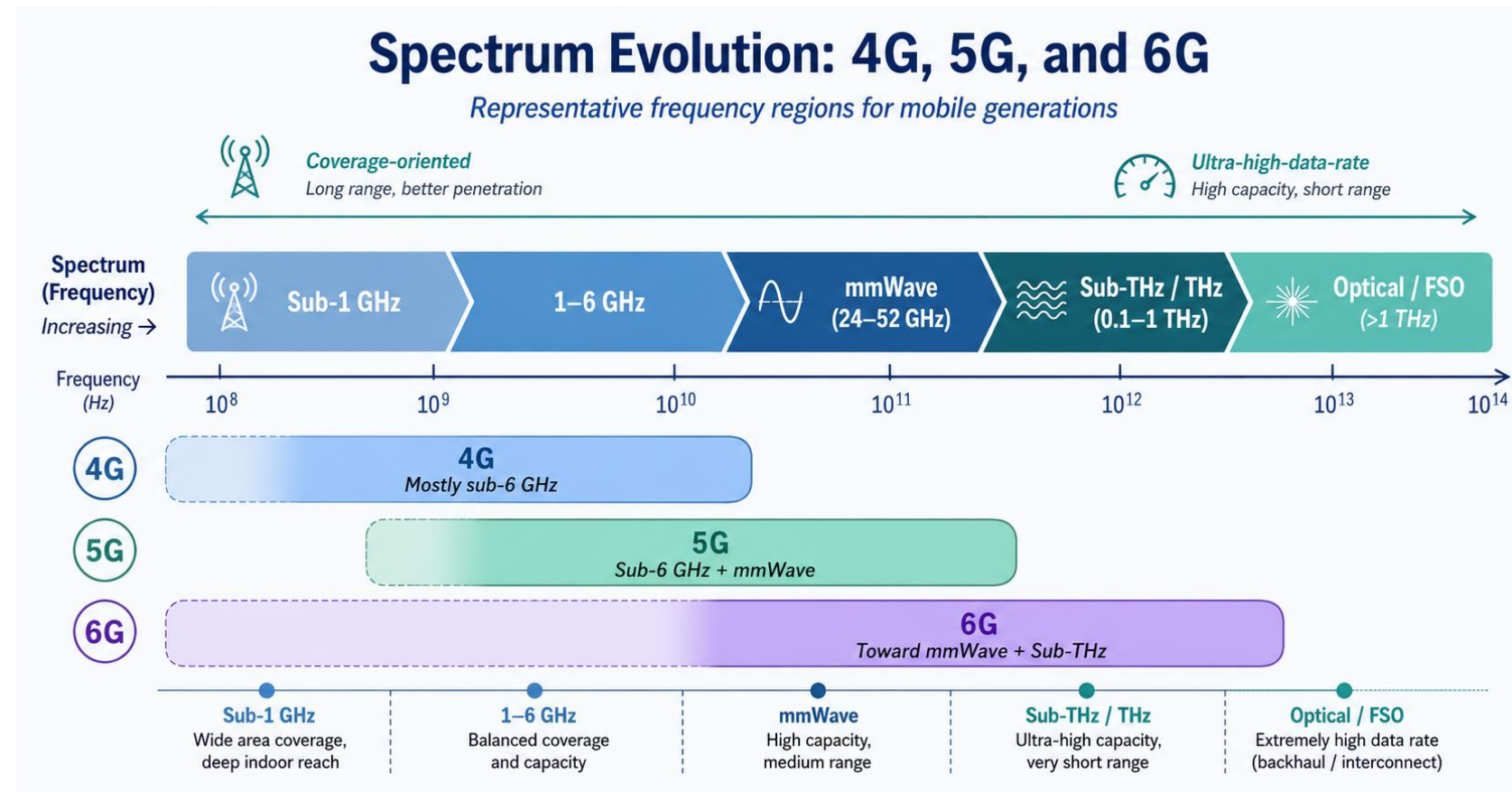
The FSO Backhaul and sub-THz access

FSO backhaul (GS-> NTN-> UAV):

- Use near-infrared light to transmit data
- Extremely high data rate (Gbps to Tbps).

Sub-THz access (UAV -> end users):

- Large bandwidth for high-rate user access (few Gbps)



The integrated FSO backhaul-based NTNs and sub-THz access can provide **large coverage** and **high-data rate** to end users.

The optical IRS-assisted FSO-based NTN

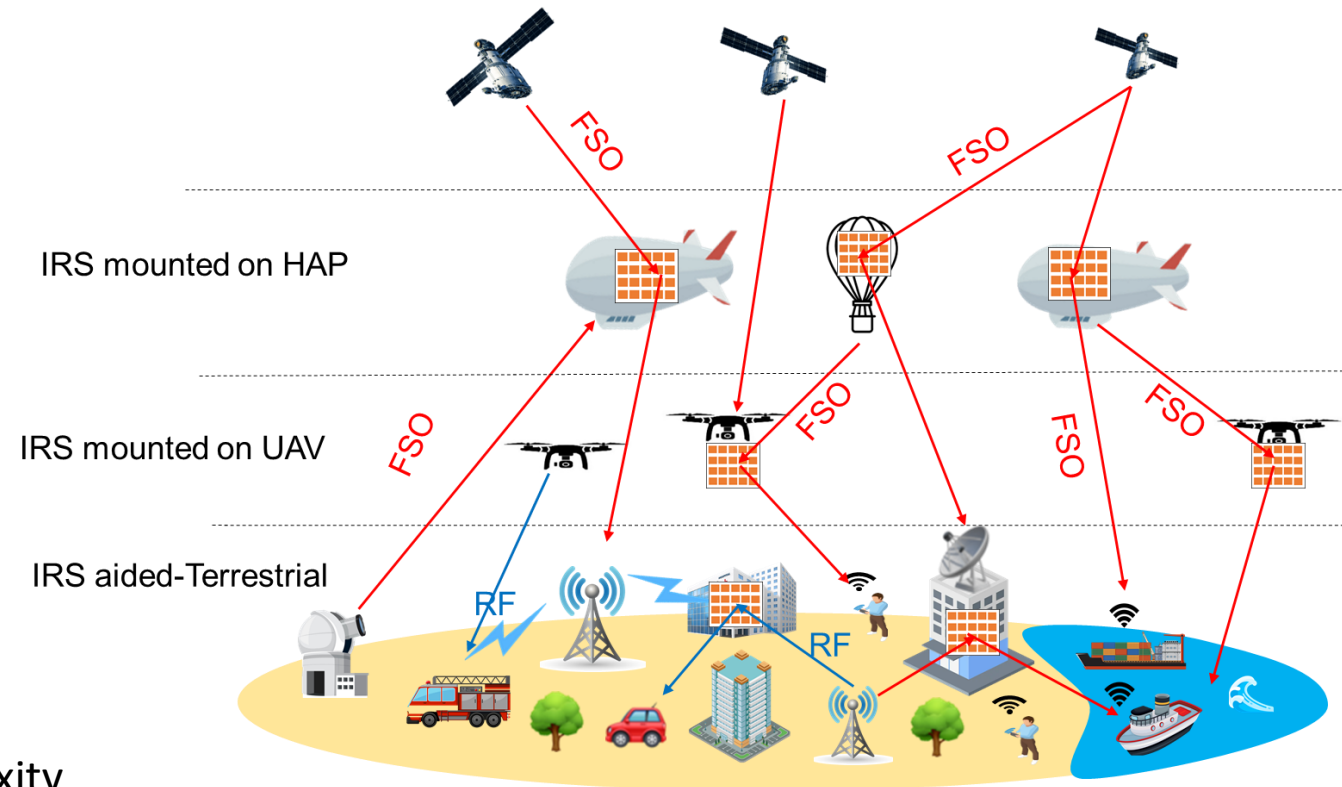
- ❑ FSO backhaul is attractive but aerial platforms have strict limits on weight, size, power consumption, and hardware complexity, since aerial platforms must carry optical components to receive, process, or transmit data.
- ❑ Many studies have developed **optical intelligent reflecting surfaces (OIRS)** as a potential technology to replace the conventional relay systems.

❖ Optical Intelligent Reflecting Surface (OIRS)

- A passive surface with mirrors or meta-atoms **reflects FSO beam** in a **controlled way**.
- Possible to simultaneously support multiple users

❖ Advantage compared to relay systems

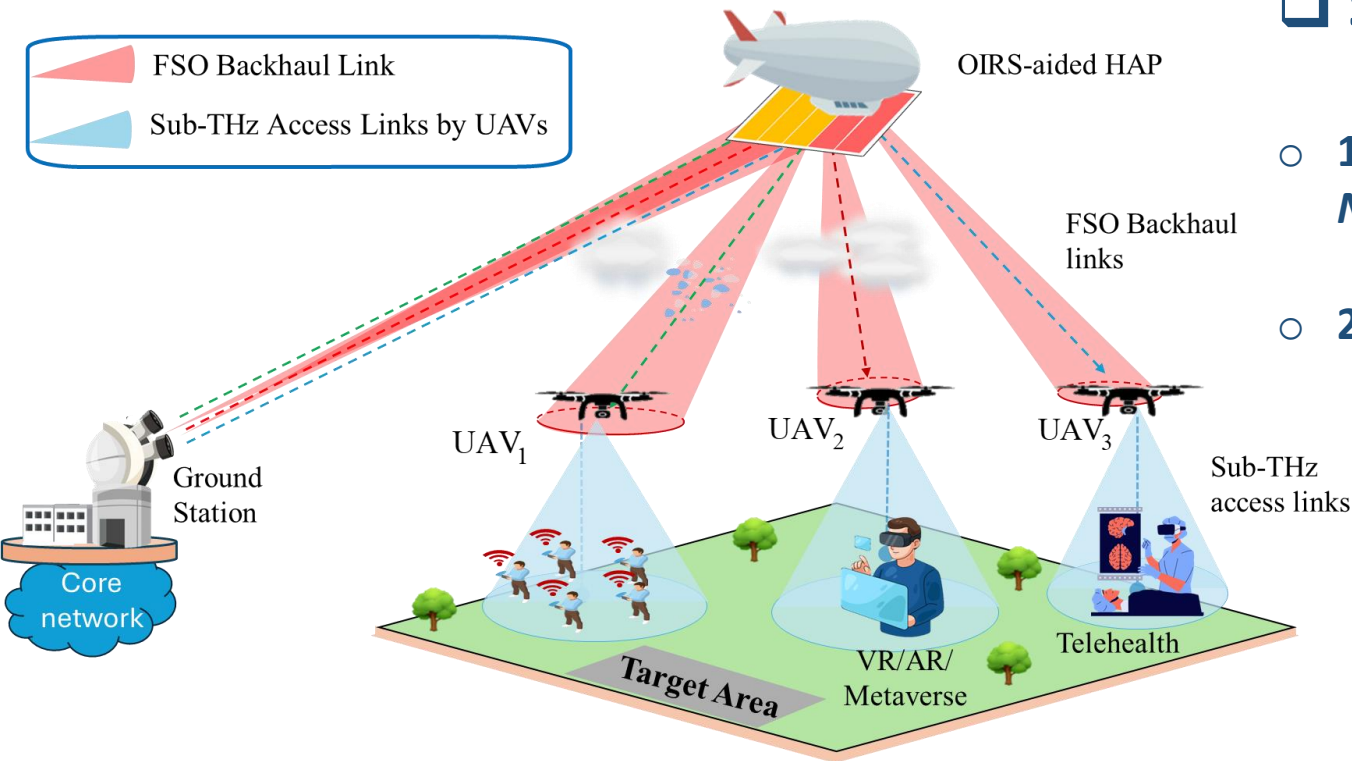
- Conventional relay (receiver-transmitter pair): to support N users $\rightarrow$ N transceivers $\rightarrow$ expensive
- OIRS approach: **1 OIRS can support N users**
 $\rightarrow$ low cost, energy consumption and reduced complexity



The operation concept of optical IRS

➡ The optical IRS mounted on HAP can reflect multiple FSO beams towards multiple UAVs by allocating IRS elements, enabling high-data rate backhaul for access networks.

The integrated FSO-based HAP and sub-THz access network



Integrated OIRS-aided HAP and sub-THz
Access Networks

System Model:

- 1. FSO Backhaul: Ground station → HAP mounted OIRS → N fixed UAVs
- 2. Sub-THz access: N UAVs → dynamic ground users.

Main challenge:

- The OIRS elements mounted on HAP is **limited**
- Traffic mismatch between **backhaul conditions** and **access demand**



***OIRS Elements Allocation
taking into account both backhaul's quality condition and access demand***

Literature Review in the Integrated Backhaul/Access Networks

- ❑ To the best of my knowledge, most of the existing studies *separately optimize* either *access optimization (1)* or *FSO/OIRS backhaul (2)* or *Integrated RF access/Backhaul (3)*, but *do not address OIRS allocation for sub-THz access*.

(1) Access networks: UAVs serve RF/mmWave users

❑ Ideas:

- UAV placement
- UAV trajectory
- Jointly UAV trajectory or UAV placement and bandwidth allocation or energy consumption

❑ Targets:

- Maximize number of supported users
- Maximize users' satisfaction, i.e., MoS
- Minimal UAV energy consumption

❑ *Assumption: backhaul is assumed sufficient, simplified, or not the bottleneck to access networks.*

- ❑ However, *in sub-THz access networks*, access demand is increasing even exceed the available FSO backhaul capacity.
=> *Therefore, backhaul may become the bottleneck, and the OIRS allocation for backhaul links becomes essential.*

(2) The OIRS-assisted FSO backhaul: OIRS serves UAVs

❑ Ideas:

- Deploy OIRS to avoid cloud, blockage
- OIRS allocation to multiple UAVs

❑ Targets:

- Channel modeling and performance evaluation for single user
- Maximize number of supported UAVs under fixed target backhaul rate

❑ *Existing OIRS allocation schemes do not address the access demand*

❑ For integrated networks, OIRS allocation **not only depend on backhaul condition** but also **access demand**

(3) The OIRS-assisted integrated FSO/RF:

- ❑ **Conventional integrated FSO/RF:** allocate OIRS equally and focus on UAV placement to maximize number of users.
 - In RF access network, under constraints of UAV coverage and limited power, access demand is much smaller than backhaul capacity -> equal OIRS allocation can still work reasonable [12].
- ❑ **Integrated FSO/sub-THz access:** OIRS allocation becomes strongly coupled with access traffic and backhaul condition because of access demand is increasing.
- ❑ Existing works does not address access-traffic aware -> Access link's satisfaction is not considered.

	FSO backhaul condition	Access demand	Interpretation	Equal OIRS Allocation [11]	Channel Based-Allocation [4]
UAV 1	Good / clear FSO link/ Maximum 10 Gbps	10 Gbps	Can efficiently use many OIRS blocks	Allocated Backhaul: 5 Gbps -> 5 Gbps is unsastified	10 Gbps
UAV 2	Poor / cloudy FSO link/ Maximum 5 Gbps	1 Gbps	Needs only a small amount of backhaul to satisfy its users	Allocated backhaul: 2 Gbps -> 1 Gbps is unused	0
				Waste of OIRS resource	May ignore UAVs

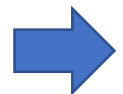
Research Gaps and Goals

Refs	System Model	Main Objective	OIRS-FSO Backhaul	Integrated Access/Backhaul	sub-THz Access	Traffic-Aware Demand Satisfaction	Moving Cloud/ Multi-HAP
UAV RF access [1]-[3]	UAV-assisted RF access	UAV placement / trajectory to maximize covered users, throughput, or user satisfaction	X	△	X	△	X
OIRS-assisted FSO backhaul [4]	OIRS-assisted FSO backhaul	Improve FSO capacity, outage, or link reliability	✓	X	X	X	△
OIRS allocation Backhaul [5]	HAP-OIRS-UAV FSO backhaul	Maximize number of supported UAVs under fixed target backhaul rate	✓	X	X	X	✓
Integrated backhaul/RF access [6]-[10]	UAV IAB with RF/mmWave access	Joint UAV placement / trajectory / bandwidth allocation	△	✓	X	△	△
OIRS-HAP + UAV RF access under clouds [11]	HAP-OIRS FSO backhaul + UAV RF access	Maximize number of supported mobile users under dynamic users/clouds	✓	✓	X	△	✓
This work	HAP-OIRS FSO backhaul + UAV sub-THz access	Maximize end-to-end traffic demand satisfaction by OIRS allocation	✓	✓	✓	✓	Future extension: ✓

✓: explicitly considered

△: partially considered /simplified

X: not considered



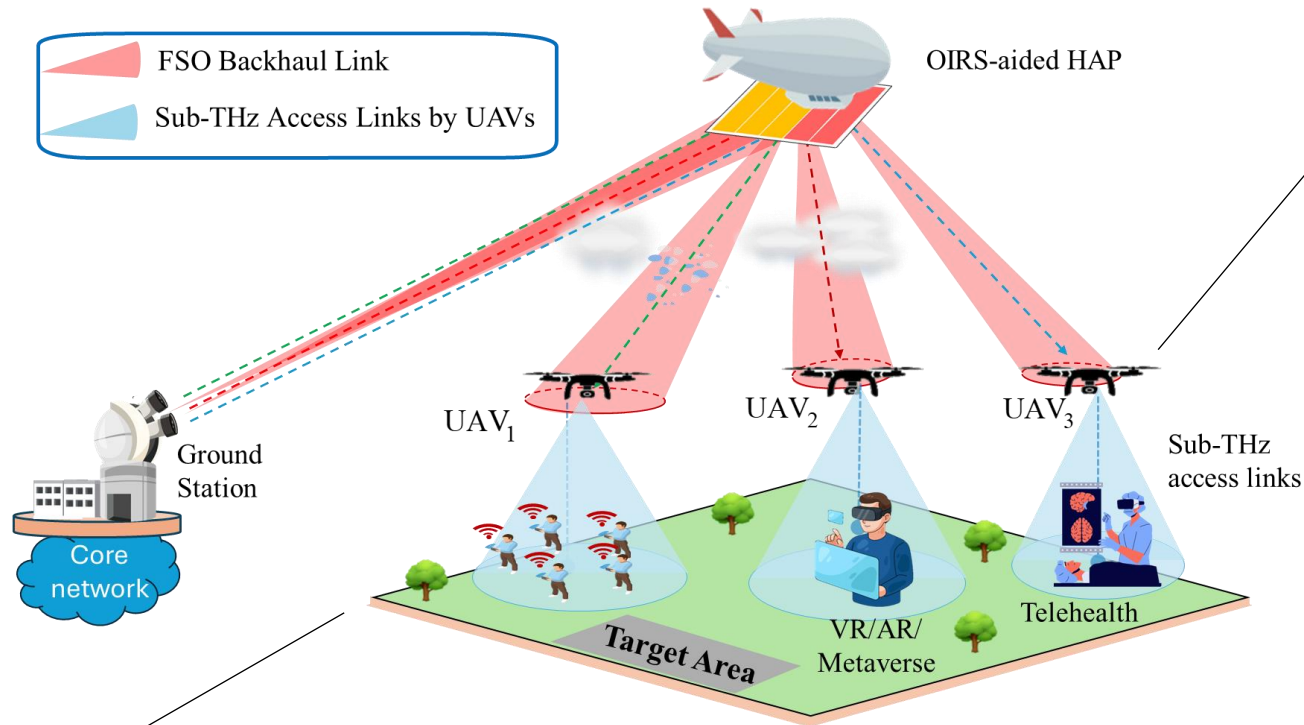
This work aims to *design a traffic-aware OIRS allocation scheme* taking into account *sub-THz access demand* and *backhaul channel condition*, in order to *maximize end-to-end traffic demand satisfaction*.

Outline of Presentation

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Access Demand-Aware OIRS Elements Allocation in Integrated FSO/sub-THz Networks



- Number of users associated with UAV_i

$$K_i(t) \sim \text{Poisson}(\lambda_i(t))$$



- Per-user requested traffic demand

$$d_{i,k}(t) \in \{d_{\text{IoT}}, d_{\text{video}}, d_{\text{eMBB}}, d_{\text{emergency}}\}$$



- Aggregate access demand of UAV _i:

$$D_i^{\text{traffic}}(t) = \sum_{k \in \mathcal{K}_i(t)} d_{i,k}(t)$$



Sub-THz access capacity of UAV_i:

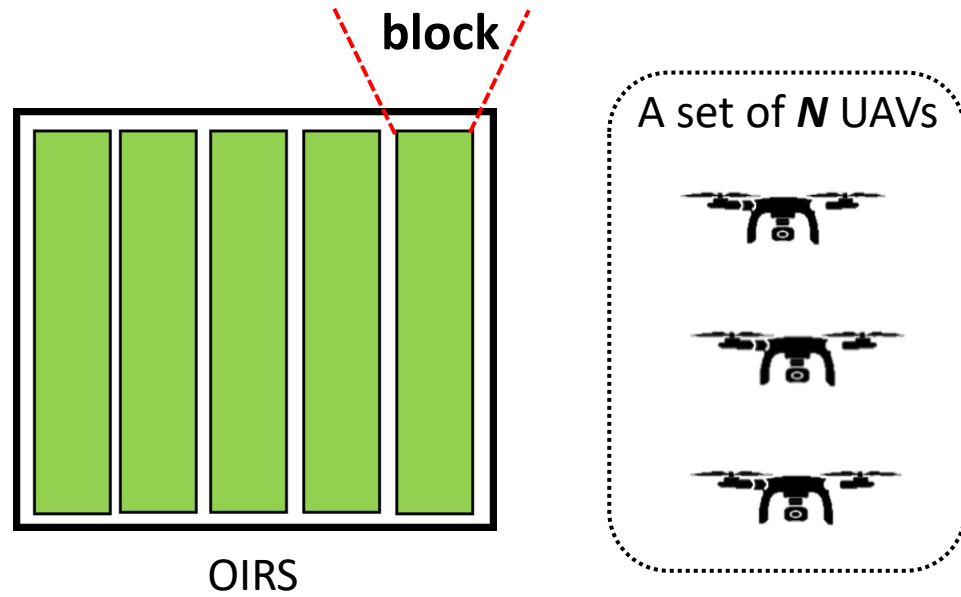
$$C_i^{\text{AC}}(t) = \sum_{k \in \mathcal{K}_i(t)} B_{\text{THz}} \log_2 (1 + \gamma_{i,k}^{\text{THz}}(t))$$

- Effective access demand requiring backhaul of UAV_i:

$$D_i(t) = \min (D_i^{\text{traffic}}(t), C_i^{\text{AC}}(t))$$



Access Demand-Aware OIRS Elements Allocation in Integrated FSO/sub-THz Networks



$$\sum_{i \in \mathcal{N}} n_i(t) \leq N_B,$$

- Access-demand satisfaction utility function:

$$U_i(n_i, t) = \log(1 + \rho_i(n_i, t)),$$

- Backhaul rate with n allocated blocks

$$R_i^{\text{BH}}(n_i; t) = B_{\text{FSO}} \log_2(1 + \gamma_i^{\text{FSO}}(n_i; t))$$



- Actual end-to-end served rate with n allocated blocks

$$r_i(n_i; t) = \min(R_i^{\text{BH}}(n_i; t), D_i(t)).$$



- Access-demand satisfaction** ratio of each UAV

$$\rho_i(n_i, t) = \frac{\text{Served rate with } n_i \text{ blocks}}{\text{Access demand}} = \frac{r_i(n_i, t)}{D_i(t) + \delta}, \quad 0 \leq \rho_i(n_i, t) \leq 1,$$



$\rho_i(n_i, t) = 1$, Access demand associated with UAV_{*i*} is fully served
 $\rho_i(n_i, t) = 0$, Access demand associated with UAV_{*i*} is not served

□ Optimization Problem:

$$\begin{aligned}
 \text{P1 : } \quad & \max_{\{n_i(t)\}} \sum_{i \in \mathcal{N}} \log(1 + \rho(n_i, t)) = \sum_{i \in \mathcal{N}} \log \left(1 + \frac{\min(R_i^{\text{BH}}(n_i; t), D_i(t))}{D_i(t) + \delta} \right) \quad \leftarrow \text{Maximize total access-demand satisfaction} \\
 \text{s.t. } \quad & \sum_{i \in \mathcal{N}} n_i(t) \leq N_B, \quad \leftarrow \text{Allocated blocks to UAV } i \\
 & n_i(t) \in \mathbb{Z}_+, \quad \forall i \in \mathcal{N}. \quad \leftarrow \text{Allocated blocks} < \text{total \# blocks}
 \end{aligned}$$

□ Algorithm: Marginal Satisfaction Utility Allocation

- Key idea: allocate each OIRS block to UAV with highest marginal satisfaction

Marginal satisfaction gain: $\Delta_i^{\text{util}}(n_i; t) = U_i(n_i + 1; t) - U_i(n_i; t)$ where $U_i(n_i; t) = \log \left(1 + \frac{\min(R_i^{\text{BH}}(n_i; t), D_i(t))}{D_i(t) + \delta} \right)$.

- By using log-function, the algorithm allocated OIRS blocks based on lowest satisfaction to highest satisfaction
- Algorithm complexity: $\mathcal{O}(N_B)$

Access Demand-Aware OIRS Elements Allocation in Integrated FSO/sub-THz Networks

	Backhaul condition	Access demand	Useful served-rate gain per OIRS block
UAV 1	Good / clear FSO link	D1= 24 Gbps	3 Gbps/block
UAV 2	Poor / cloudy FSO link	D2 =1 Gbps	0.6 Gbps/blocks

Number of OIRS blocks: $N_B = 8$

Method	Allocation criterion	Strategy in this example	Final allocation	Result
Equal allocation	All UAVs receive the same OIRS blocks	Equally allocate OIRS to N UAVs	$n=[4,4]$	UAV1: (12/24) Gbps, ($p_1=0.5$). UAV2: (1/1) Gbps, ($p_2=1$). Some OIRS blocks are wasted on UAV2. Total satisfaction ratio: 1.5
Total Rate / channel-based allocation	Allocate OIRS blocks to UAV with the largest marginal served-rate gain	Prioritize allocate UAV1, then UAV2	$n=[8,0]$	UAV1: (24/24) Gbps, ($p_1=1$). UAV2: (0/1) Gbps, ($p_2=0$). UAV2 is ignored, although its demand is small. Total satisfaction ratio: 1
Proposed scheme	Allocate OIRS block to the largest marginal satisfaction utility gain	First allocate a small number of blocks to UAV2 to satisfy its low demand, then allocate the remaining blocks to UAV1.	$n=[6,2]$	UAV1: (18/24) Gbps, ($p_1=0.75$). UAV2: (1/1) Gbps, ($p_2=1$). The minimum satisfaction is improved and no UAV is completely ignored. Total satisfaction ratio: 1.75

Performance Metrics and Benchmarks for Comparison

□ Performance metrics:

- Total satisfaction utility: $U_{\text{total}}(t) = \sum_{i \in \mathcal{N}} \log(1 + \rho_i(t))$
- Average demand satisfaction ratio: $\bar{\rho}(t) = \frac{1}{N} \sum_i \rho_i(t)$
- Total served rate: $R_{\text{total}}(t) = \sum_i r_i(t)$
- Minimum demand satisfaction: $\rho_{\min}(t) = \min_i \rho_i(t)$

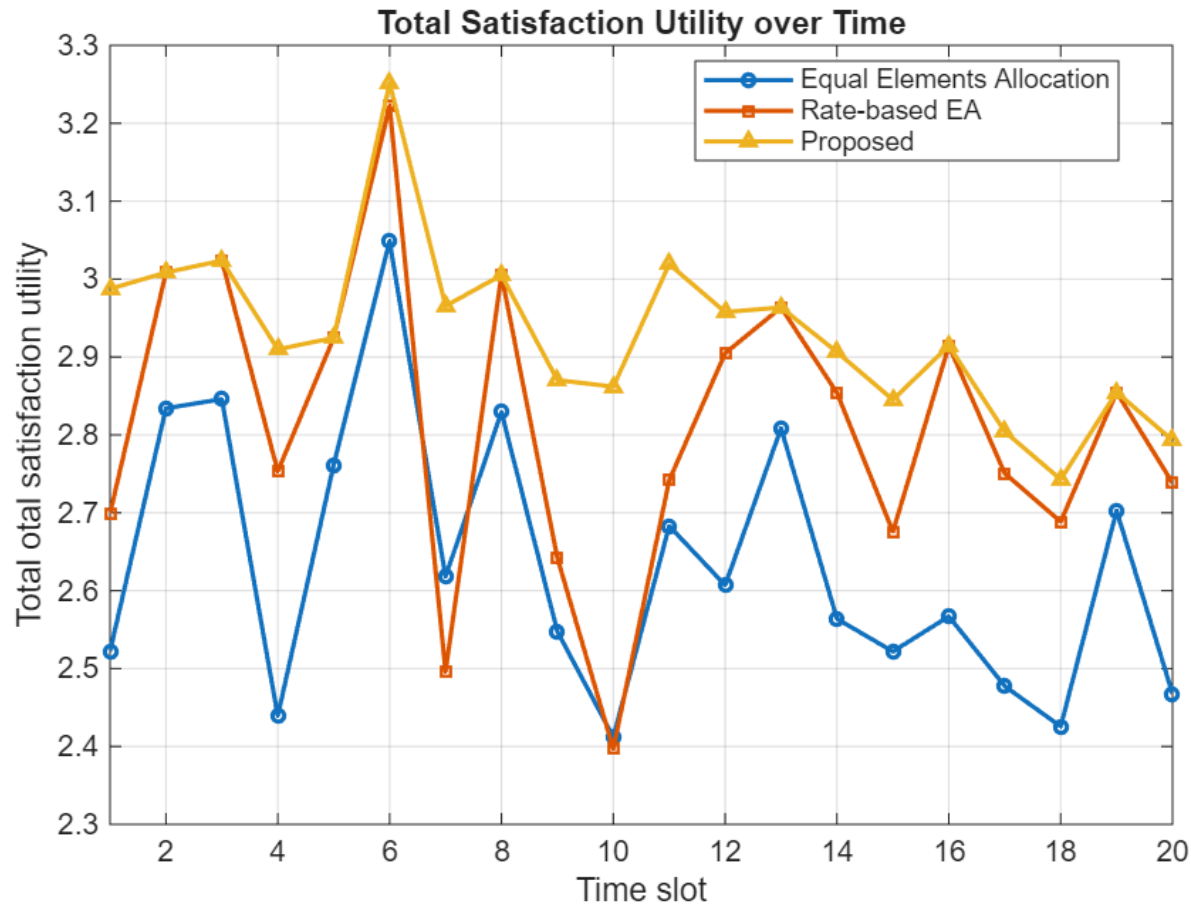
□ To evaluate our proposed scheme, we compare it with two benchmarks:

- (1) **Rate-based Element Allocation (REA)**: allocate OIRS based on highest marginal rate gain
- (2) **Uniform Element Allocation (UEA)**: the OIRS is shared equally among all users from the start.

□ Parameters: 5 UAVs with [15, 5, 12, 8, 10] users, respectively.

- Service rate classes of users = [100 Mbps, 600Mbps, 1Gbps, 2Gbps];
- Cloud liquid water content = [0.5, 2, 1.5, 1.5, 0.5];

The Effectiveness of the Proposed Access Demand-Aware OIRS Allocation

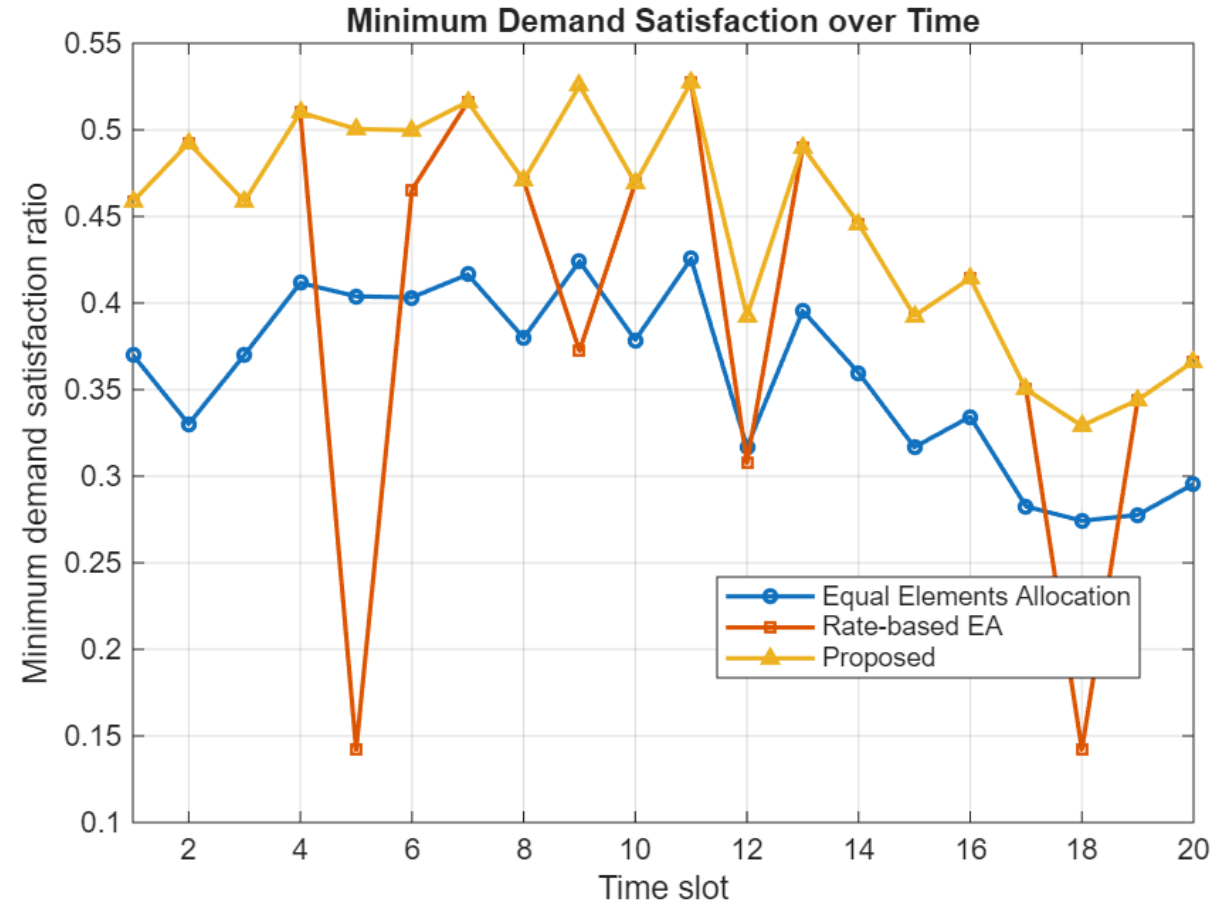
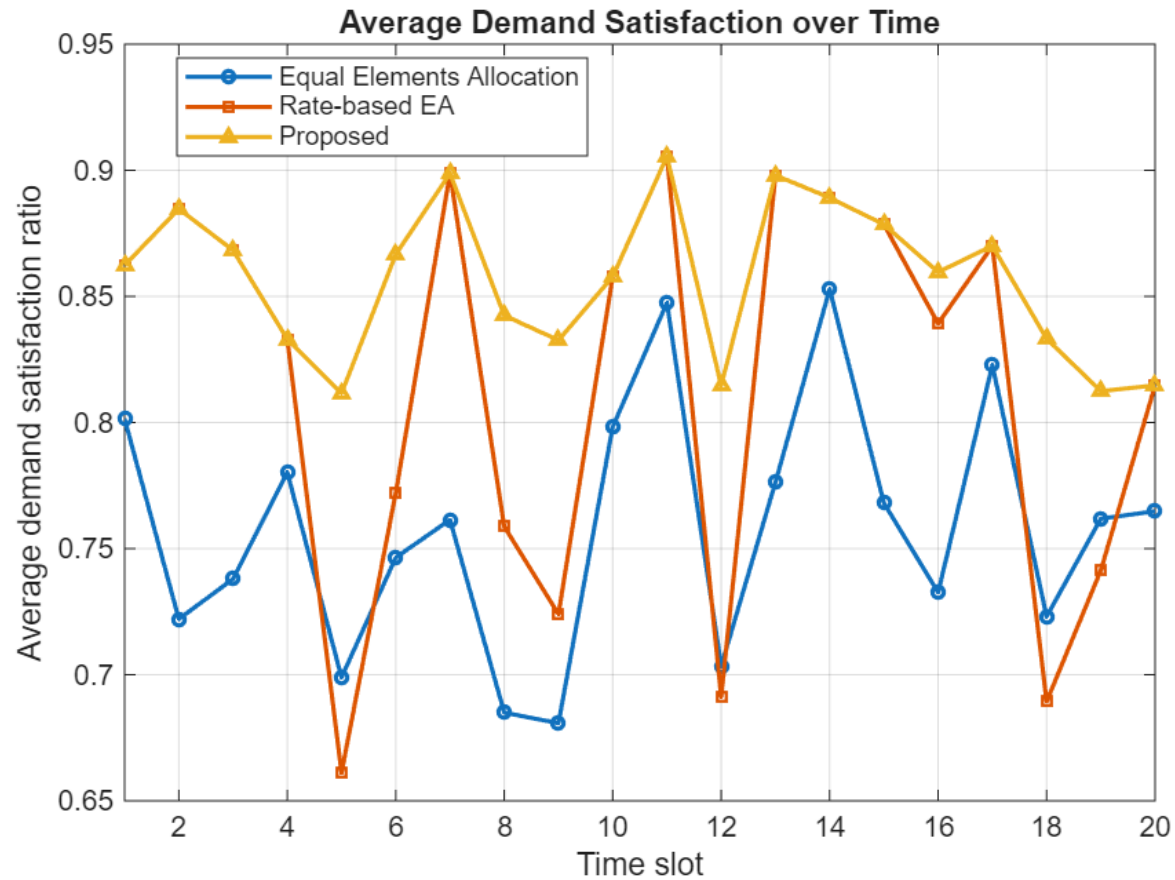


*maximum satisfaction ratio = $N \cdot \log(2) = 3.465$



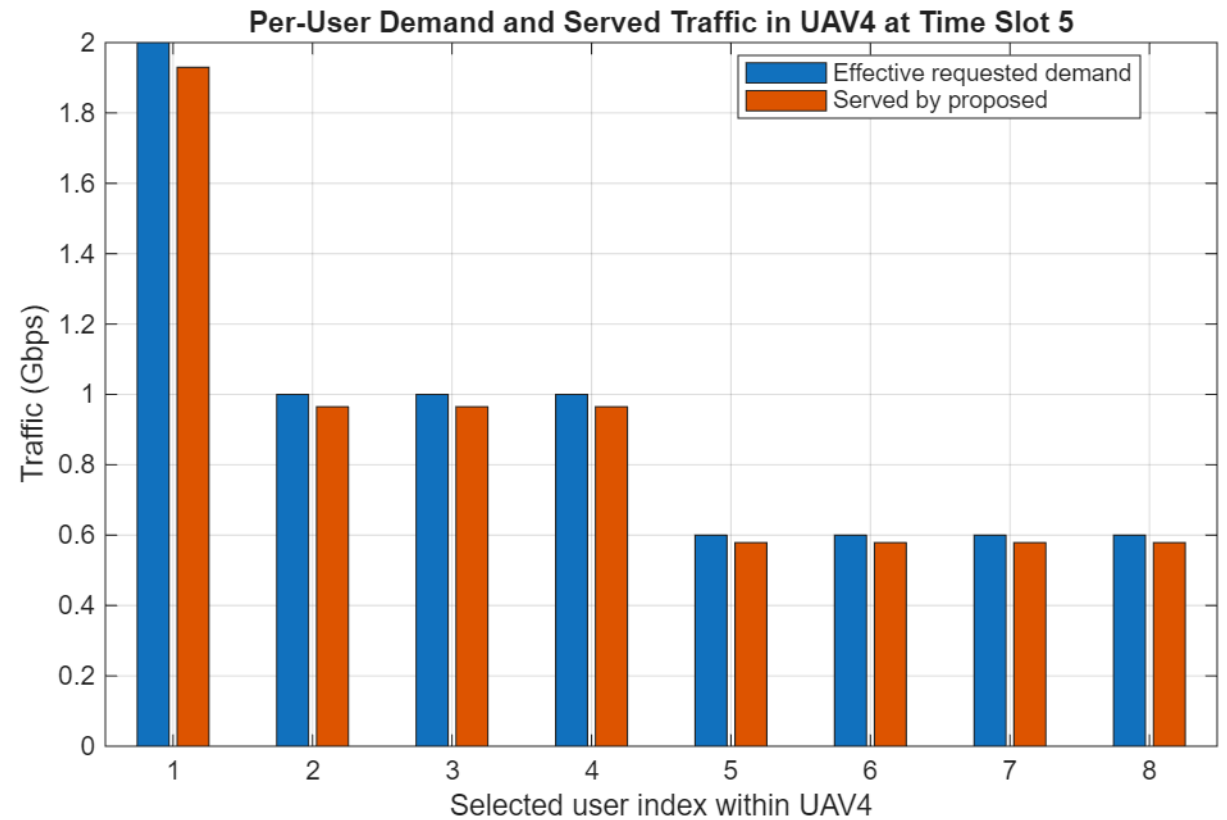
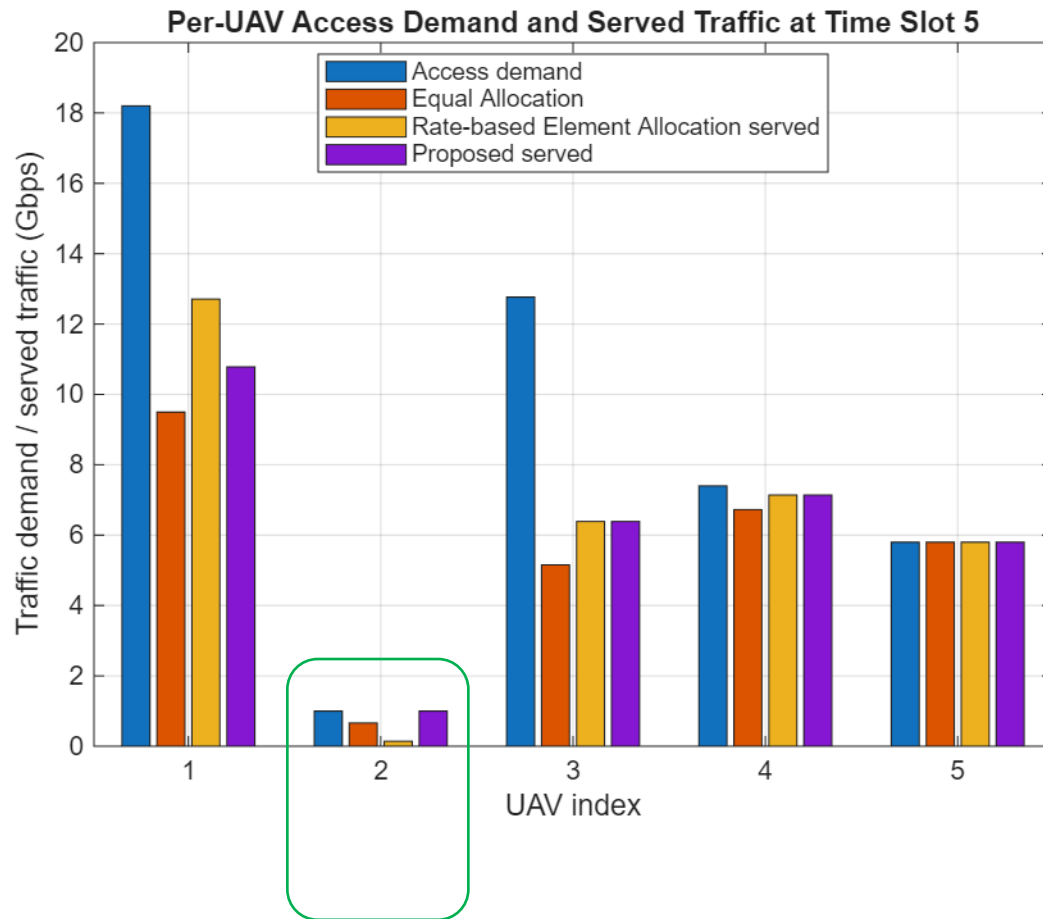
- The proposed allocation **outperforms other schemes** by **improving total access-demand satisfaction** while the **total rate is comparable** to rate-optimized scheme

The Effectiveness of the Proposed Access Demand-Aware OIRS Allocation



- The proposed scheme *provide higher average access-demand satisfaction* and *guarantee higher minimum demand satisfaction*.

The Effectiveness of the Proposed Access Demand-Aware OIRS Allocation



- The proposed scheme *guarantees to allocate to bad UAV -> improve user's satisfaction*

- ❑ In this work, we proposed a access demand-aware OIRS allocation for integrated FSO-based HAP and sub-THz access networks.
 - The proposed scheme **effectively allocates** OIRS elements to **maximizes the total access-demand satisfaction**.
 - Numerical results confirmed the **effectiveness of proposed scheme** compared to conventional schemes.
- ❑ **Future extension: Jointly traffic-aware element allocation and HAP switching under moving cloud.**
 - Multi-HAP extension: recent works focus on deploying multi-HAP to avoid cloud, improving outage probability
=> HAP selection, HAP optimal position and OIRS allocation under realistic cloud map
 - Moving cloud can lead to HAP switching and OIRS allocation change fast
=> Jointly HAP switching and OIRS element allocation algorithm under moving cloud

THANK YOU FOR LISTENING!

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